Music Recommendation System

November 2025, prefer time slot: 12/29 (the third week)

1 Introduction

We would like to build a music recommendation system. Listeners are able to use this system by typing tags, styles, etc. to search for their favorite music, and our system will calculate the similarity of those music.

2 Methods

In this project, we plan to try out and compare three common types of recommendation methods. The goal is to see how each method behaves and which one works better on the music data.

1. Content-Based Recommendation

We will represent each song using both audio and textual features. Audio features include tempo, loudness, and timbre, while textual features come from metadata such as artist, genre, and tags. Similarity will be computed using cosine similarity for audio and TF-IDF or Jaccard similarity for text. By combining these scores, the system can recommend songs that are similar in both sound and description, even without user ratings.

2. Collaborative Filtering

We will test both user-user and item-item collaborative filtering. The idea is to find users or items that are similar based on their rating patterns. Predictions will be made by taking the weighted average from the k nearest neighbors.

3. Latent Factor Model (Matrix Factorization)

We will also try a simple matrix factorization model, which learns hidden features for users and items. The predicted rating comes from the interaction of these hidden vectors and some basic bias terms. We will train the model using stochastic gradient descent with regularization.

3 Expected Outcomes

- Implement the three methods: content-based, collaborative filtering, and matrix factorization.
- Compare their results and observe how each method performs on the same dataset.
- Summarize the pros and cons of each method based on the experimental results.

4 Dataset

We will use the **Million Song Dataset** and its 10k subset for our experiments. URL: `http://millionsongdataset.com/`

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